KNIME POC: Node-by-node guide

Workflow: **KNIME_Alteryx_POC_Sales_Benchmark** (20 nodes) · Prepared by Arjun for the Tax team demo

Input: orders_10k.csv, products.csv, customers.csv

Output: data/output/knime_poc_results.xlsx (3 sheets)

Row counts below: 10K orders file, calculated independently; confirm on your own run

1 Ingestion

1 CSV Reader (orders)

- Reads `orders_10k.csv` from the workflow's own `data` folder, using a path relative to the workflow. Because the path is relative, the workflow is portable: copy it to another machine and it still finds its data.
- The column types are fixed in the node's settings: `quantity` and `discount_pct` as Integer, everything else as String. Fixed types (rather than auto-detected ones) mean a bad input file fails loudly instead of being silently misread.
- **Output: 10,046 rows × 7 columns.**

2 CSV Reader (products)

- Reads the product master: 500 rows with `product_id`, `product_name`, `category` and `unit_price` (Double).
- This is a reference (lookup) table that enriches the orders later on.

3 CSV Reader (customers)

- Reads the customer master: 10,000 rows.
- The reader trims leading and trailing spaces automatically, which already fixes the padded region values (for example `" North "`).

2 Data quality and cleansing

4 Duplicate Row Filter

- Removes duplicate rows, comparing all 7 columns, so only rows that are identical in every field count as duplicates.
- It keeps the first occurrence and preserves the original row order.
- **Output: 10,000 rows (46 duplicates removed).**
- *Talking point: in tax data, this is the step that catches an invoice extracted twice.*

5 String Manipulation (build ISO date)

- Adds a helper column, `date_dmy_fixed`, by rearranging a `dd/MM/yyyy` string into `yyyy-MM-dd`.
- Expression:

```
join(substr($order_date$,6,4), "-", substr($order_date$,3,2), "-",
substr($order_date$,0,2))
```

6 Rule Engine (normalise order_date)

- Applies conditional logic to the date column, overwriting it with the result:

```
$order_date$ LIKE "*/*" => $date_dmy_fixed$
TRUE => $order_date$
```

  If the date has slashes, use the converted value; otherwise keep it as it is.
- Rules are checked in order, top to bottom, and the first match wins.
- After this node, every date is in one standard format (ISO 8601). About 10% of rows were in the other format.
- *Talking point: this is the multi-node pattern we discussed. KNIME splits string functions and conditional logic across separate nodes, where Alteryx would do it in one Formula.*

7 String Manipulation (derive order_month)

- Adds `order_month` = `substr($order_date$, 0, 7)`, which gives `yyyy-MM` for the monthly trend.

8 Rule-based Row Filter (missing quantity)

- Filters rows using a rule:

```
MISSING $quantity$ => FALSE
TRUE => TRUE
```

  Rows with a blank quantity are dropped; everything else is kept.
- **Output: 9,806 rows (194 removed).**
- *Talking point: this is an explicit rule for null values. A production version would send these rows to an exceptions file instead of dropping them.*

9 Rule-based Row Filter (status)

- Keeps only rows where `$status$ = "Completed"`, which excludes cancelled and returned orders.
- **Output: 7,819 rows.**

10 String Manipulation (clean region)

- Applies `capitalize(lowerCase(strip($region$)))` to the customer table, replacing the original column.
- It standardises `NORTH`, `north` and `" North"` to `North`, so the later grouping doesn't split one region into several groups.
- *Talking point: the same technique standardises vendor names or state codes in tax data.*

3 Integration (joins)

11 Joiner (orders ⋈ products)

- An inner join on `product_id`, with the many orders matched to one product each.
- It brings in `product_name`, `category` and `unit_price`, and drops the helper column `date_dmy_fixed`.
- Matching is exact on value and data type.
- *Talking point: the Joiner can also send unmatched rows from either side to separate output ports. In a reconciliation, those ports are your exceptions report: records in the books but not in GSTR-2B, and the other way round.*

12 Joiner (+ customers)

- An inner join on `customer_id` that adds the customer attributes (name, city, region, segment).
- **Output: 7,819 rows, the fully enriched order-level dataset. The row count doesn't change, which confirms every order found its product and its customer.**

4 Business logic

13 Math Formula (revenue)

- Adds `revenue = $quantity$ * $unit_price$ * (1 - $discount_pct$ / 100)`.
- It computes row by row, and the result is a Double.

14 Number Rounder

- Rounds `revenue` to 2 decimal places using HALF_UP rounding (standard financial rounding).
- Rounding each row before aggregating (rather than rounding the totals afterwards) is what makes the totals reconcile exactly with the independent calculation.
- *Talking point: this is the same rounding question that matters for tax computations.*

5 Aggregation and output

15 GroupBy (region × category)

- Groups on `region` and `category`, and aggregates `Count(order_id)`, `Sum(quantity)` and `Sum(revenue)`.
- **Output: 25 rows (5 regions × 5 categories).**

16 Sorter

- Sorts the summary by `Sum(revenue)`, highest first.

17 GroupBy (monthly trend)

- Groups on `order_month`, giving the order count and revenue for each month (24 months).

18 Sorter

- Sorts the months in ascending order.

19 GroupBy (reconciliation totals)

- Has no group columns, so it aggregates the whole table into a single control-total row.
- **Expected: 7,819 orders and revenue of 705,199,819.82.**
- *Talking point: this is the control total. In audit terms, it's how we prove the output is complete and accurate against the source.*

20 Excel Writer

- Writes all three result tables to one workbook, `data/output/knime_poc_results.xlsx`, one sheet each: `summary_region_category`, `monthly_trend` and `reconciliation_totals`.
- It overwrites the file on each run, so re-running is safe and repeatable.

✓ Row-count checkpoints

After node	Rows	Change
1 · CSV Reader (orders)	10,046	Raw input
4 · Duplicate Row Filter	10,000	−46 duplicates
8 · Missing quantity filter	9,806	−194 blanks
9 · Status filter	7,819	Completed only
12 · Second Joiner	7,819	No loss on joins
19 · Reconciliation totals	1	Revenue 705,199,819.82

i General points for the demo

- **Traffic lights:** red means not configured, amber means configured and ready, green means executed.
- **Inspect any node:** right-click → output table to see the data at that step. This is the debugging and audit-trail advantage.
- **Partial re-run:** change one node and only that node and the nodes after it re-execute.
- **Re-use:** change the input file in node 1, run it again, and get the same logic on new data.
- **Be upfront about the legacy nodes:** if the team asks why some dialogs open in the older style, say: “Nodes 5–10, 13 and 14 are classic nodes. KNIME’s newer Expression node can replace them in fewer steps, and that’s the next iteration.”